

HW 14 Fourier Series

Problem 1

$$\left. \begin{aligned} \text{(i)} \quad \int \sin(nx) dx &= -\frac{1}{n} \cos(nx) + C \\ \text{(ii)} \quad \int \cos(nx) dx &= \frac{1}{n} \sin(nx) + C \end{aligned} \right\} \begin{array}{l} \text{follows by } u\text{-sub} \\ \text{with } u=nx \end{array}$$

$$\text{(iii)} \quad \int x \sin(nx) dx$$

Integration by Parts

$$= -\frac{x}{n} \cos(nx) + \int \frac{1}{n} \cos(nx) dx$$

$$\begin{array}{ll} u=x & v=-\frac{1}{n} \cos(nx) \\ du=dx & dv=\sin(nx) dx \end{array}$$

$$= -\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) + C$$

$$\text{(iv)} \quad \int x \cos(nx) dx$$

$$\begin{array}{ll} u=x & v=\frac{1}{n} \sin(nx) \\ du=dx & dv=\cos(nx) dx \end{array}$$

$$= \frac{x}{n} \sin(nx) - \frac{1}{n} \int \sin(nx) dx$$

$$= \frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) + C$$

Problem 2

$$\text{(i)} \quad \sin(n\pi) = 0 \quad \forall n \in \mathbb{N}$$

$$\text{(ii)} \quad \sin(2n\pi) = 0 \quad \forall n \in \mathbb{N}$$

$$\text{(iii)} \quad \sin(3n\pi) = 0 \quad \forall n \in \mathbb{N}$$

$$\text{(iv)} \quad \sin\left(\frac{1}{2}n\pi\right) = \begin{cases} 1 & n=1, 5, 9, \dots \text{ (Every other odd starting w/ 1)} \\ 0 & n \text{ even} \\ -1 & n=3, 7, 11, \dots \text{ (Every other odd starting w/ 3)} \end{cases}$$

$$\text{(v)} \quad \cos(n\pi) = \begin{cases} 1 & n \text{ even} \\ -1 & n \text{ odd} \end{cases} \quad \text{or} \quad (-1)^n$$

$$(vi) \cos(2n\pi) = 1 \quad \forall n \in \mathbb{N}$$

$$(vii) \cos(3n\pi) = \begin{cases} 1 & n \text{ even} \\ -1 & n \text{ odd} \end{cases} \quad \text{or} \quad (-1)^n$$

$$(viii) \cos\left(\frac{1}{2}n\pi\right) = \begin{cases} 0 & n \text{ odd} \\ -1 & n = 2, 6, 10, \dots \text{ (Every other even starting w/2)} \\ 1 & n = 4, 8, 12, \dots \text{ (Every other even starting w/4)} \end{cases}$$